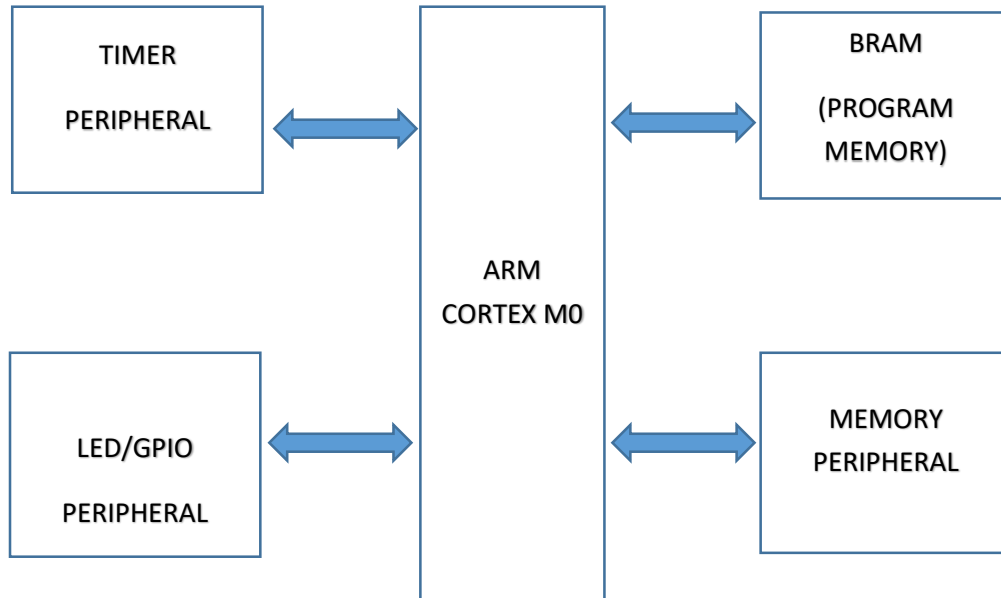


MINI PROJECT



Design and develop an ARM Cortex M0 based SoC by interfacing the peripherals Timer, Memory (255x32 Single port RAM) and GPIO/LED.

The SoC should work as follows:

- 1)** Timer should count from '0F' to '00'. Once the count reaches the minimum value ('00'), an interrupt is generated and that should be cleared (using clear register).
- 2)** Once timer interrupt becomes high,
 - a. LED peripheral should output the value 'FF' .
 - b. Single port RAM should be written with data '55' in its first location.
 - c. After a small delay, LED peripheral should be cleared and the first location of single port RAM should be written with a value '00'.